

SHIP VESSEL CLASSIFICATION USING ACOUSTIC SIGNALS

ARJUN A A R, ¹JEFF JACOB GEORGE, ¹PRASANNA BHAVAN, ¹SANJAY SIDDARTH S ¹

¹Indian Institute of Technology Madras, Chennai, India

ABSTRACT

Passive acoustic sensing offers an effective alternative to radar and optical systems for underwater vessel identification, particularly under adverse environmental conditions. This project investigates machine learning pipelines for vessel classification using underwater acoustic recordings from the DeepShip and Qiandao22 datasets, which exhibit severe class imbalance across five vessel categories.

Three progressively more sophisticated approaches are evaluated: a classical MFCC-based Logistic Regression baseline, a CNN-based log-mel spectrogram classifier, and a sequential two-stage pipeline with a dedicated large-vessel discrimination model. Class imbalance — particularly the severe under-representation of the tanker category — was a central challenge throughout, and several mitigation strategies were explored. Additionally, several large vessel categories exhibit substantial similarity in low-frequency acoustic structure, further complicating fine-grained classification. Mitigation strategies including focal loss, weighted random sampling, and augmentation are investigated. A balanced reference dataset, DS3500, is used as a benchmark for evaluating performance under balanced training conditions.

Results show that tanker recall remains consistently low across all models despite mitigation efforts, while UUV and speedboat classification remain highly reliable despite comparatively limited support. Embedding visualisation further reveals substantial overlap between tanker, cargo ship, and passenger ship acoustic representations, suggesting that classification difficulty arises from both severe imbalance and intrinsic acoustic similarity among large commercial vessels. A macro F1-score of 0.86 on DS3500 confirms that the proposed architectures are capable of strong performance under balanced conditions.

1. INTRODUCTION

Maritime surveillance plays a critical role in naval defense, underwater security, environmental monitoring, and harbor management. Conventional vessel detection systems such as radar and optical cameras are often affected by weather conditions, fog, darkness, and oceanic disturbances. Passive acoustic monitoring offers an alternative solution by utilizing underwater radiated noise produced by ships.

Ships generate characteristic acoustic signatures due to engines, propellers, cavitation, and onboard machinery. These acoustic signals can be captured using hydrophones and analyzed using machine learning techniques for vessel classification.

Compared to radar systems, passive acoustic monitoring offers several advantages:

- Stealthy underwater sensing
- Low energy consumption
- Long-range underwater detection
- Operation under poor visibility conditions

The objective of this project is to investigate machine learning approaches for underwater vessel classification using publicly available underwater acoustic datasets.

2. PROBLEM STATEMENT

Underwater ship identification becomes challenging when environmental visibility is poor or when active radar systems are unavailable. Passive acoustic sensing enables silent vessel detection without emitting detectable signals.

The primary objectives of this work are:

- Extract discriminative acoustic features from underwater recordings.
- Develop machine learning models for vessel classification.
- Investigate both classical and deep learning approaches.
- Analyze the impact of class imbalance on classification performance.
- Study the tradeoff between handcrafted feature engineering and deep representation learning.

3. DATASETS USED

Three publicly available underwater acoustic datasets were used.

3.1. DeepShip Dataset

The DeepShip dataset contains underwater acoustic recordings of large vessels collected in deep-ocean environments.

3.2. Qiandao22 Dataset

Qiandao22 consists of underwater acoustic recordings collected in lake environments using hydrophone arrays, covering a range of small and medium vessel types operating in inland waterways.

3.3. DS3500 Dataset

DS3500 is a preprocessed and class-balanced collection of underwater acoustic recordings based on the ShipsEar dataset. It contains five vessel categories with equalized class distributions, making it well-suited as a controlled reference for evaluating classification performance independent of data imbalance effects.

The primary experimental dataset used in this work is a subset drawn from DeepShip and Qiandao22, comprising five vessel categories with severe class imbalance — particularly for the tanker class. DS3500 is used as a secondary reference experiment.

All recordings were stored in WAV audio format and treated as time-series acoustic signals.

4. METHODOLOGY

The overall machine learning pipeline is shown below:

Raw Audio → Feature Extraction → Model Training → Evaluation

4.1. Audio Preprocessing

Audio recordings were resampled to 16 kHz, silence-trimmed, and segmented into fixed-length windows. For the classical baseline, 40 MFCC coefficients were extracted per segment and summarised using per-coefficient mean, standard deviation, maximum, and minimum, yielding a 160-dimensional feature vector. For deep learning models, log-mel spectrograms were computed and stored as memory-mapped arrays. All splits followed a 70/15/15 file-level stratified partition to prevent segment leakage across splits.

4.2. Reference Experiment

A reference experiment was conducted on the DS3500 dataset to validate the effectiveness of the proposed acoustic classification pipeline. Raw audio recordings were converted into log-Mel spectrograms and a ResNet-18 model initialized with pretrained weights was then fine-tuned for five-class vessel classification. This approach achieved an accuracy of 85% on the DS3500 dataset, demonstrating that the feature extraction and classification framework can effectively learn discriminative vessel characteristics under relatively balanced conditions. The result serves as a performance reference and motivates the subsequent investigation of class imbalance mitigation strategies on the primary dataset.

4.3. Feature Extraction

Two major feature extraction approaches were investigated:

- Mel-Frequency Cepstral Coefficients (MFCCs)
- Log-Mel Spectrograms

MFCCs capture spectral envelope characteristics and are widely used in speech and audio recognition tasks.

The Fourier transform used in spectral analysis is represented as:

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N} \quad (1)$$

4.4. Handling Class Imbalance

A major challenge encountered during underwater vessel classification was dataset imbalance, where certain vessel categories contained significantly fewer samples than others. This imbalance caused the models to become biased toward majority classes, reducing classification performance for minority vessel categories. To address this issue, techniques such as focal loss, weighted random sampling, and data augmentation were incorporated during training.

4.4.1. Focal Loss Focal Loss was investigated as an alternative to standard cross-entropy loss for improving minority-class learning. Unlike conventional loss functions that treat all samples equally, focal loss reduces the contribution of easily classified samples and focuses more on difficult or misclassified examples. The focal loss function is defined as:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

where p_t represents the predicted probability for the true class, α_t is a balancing parameter, and γ controls the emphasis placed on difficult samples. By assigning greater importance to hard examples, focal loss improves robustness against class imbalance and enhances the learning of discriminative acoustic features.

4.4.2. Weighted Random Sampling and Data Augmentation Weighted Random Sampling was used during mini-batch generation to ensure that minority-class samples appeared more frequently during training. Sampling probabilities were assigned inversely proportional to class frequencies, thereby reducing model bias toward dominant vessel categories.

To further improve model generalization, data augmentation techniques were applied to acoustic recordings and spectrogram representations. Transformations such as temporal shifting, noise injection, pitch variation, time stretching, and spectrogram masking introduced realistic acoustic variability into the training data. These augmentations helped reduce overfitting and improved the model's ability to generalize to unseen underwater acoustic conditions.

4.5. Evaluation Metrics

Model performance was evaluated using multiple classification metrics to obtain a balanced assessment under imbalanced dataset conditions.

4.5.1. Precision Precision measures the proportion of correctly predicted vessel samples among all samples predicted for a particular class:

$$\text{Precision} = \frac{TP}{TP + FP}$$

where TP denotes True Positives and FP denotes False Positives. High precision indicates a low false-positive rate.

4.5.2. Recall Recall measures the proportion of actual vessel samples that were correctly identified by the model:

$$\text{Recall} = \frac{TP}{TP + FN}$$

where FN represents False Negatives. High recall is particularly important in surveillance applications where missed detections are undesirable.

4.5.3. F1-Score The F1-score combines precision and recall using their harmonic mean:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

This metric provides a balanced evaluation when both false positives and false negatives are important.

4.5.4. Macro F1-Score Macro F1-score computes the F1-score independently for each class and then averages them equally:

$$\text{Macro F1} = \frac{1}{N} \sum_{i=1}^N F1_i$$

where N is the number of classes. Since all classes are weighted equally, Macro F1-score is useful for evaluating minority-class performance.

4.5.5. Weighted F1-Score Weighted F1-score computes the class-wise F1-score while weighting each class according to its sample count:

$$\text{Weighted F1} = \frac{\sum_{i=1}^N w_i F1_i}{\sum_{i=1}^N w_i}$$

where w_i represents the number of samples in class i . This metric provides a more realistic estimate of overall performance for imbalanced datasets.

4.6. Log-Mel Spectrogram Visualization

Log-mel spectrograms were generated from underwater acoustic recordings to preserve both temporal and frequency-domain information. These spectrograms were used as input representations for CNN-based classification models. Compared to raw waveforms, spectrograms provide a more informative visualization of vessel-specific harmonic structures and acoustic signatures.

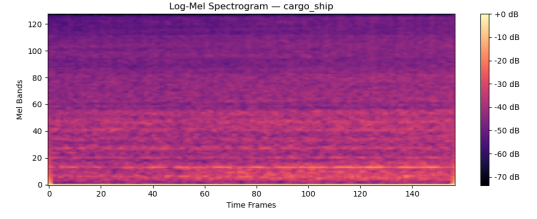


FIGURE 1: Representative log-mel spectrograms generated from underwater acoustic recordings. The spectrograms capture temporal-frequency energy distributions used as CNN input representations.

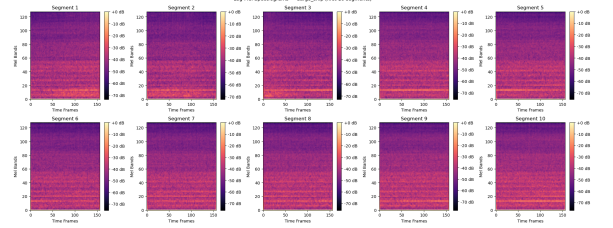


FIGURE 2: Log-mel spectrogram representations for multiple cargo ship acoustic samples. Distinct harmonic and spectral patterns corresponding to vessel noise characteristics can be observed across recordings.

The spectrogram representations reveal distinguishable frequency-energy distributions across vessel categories. Cargo ship recordings exhibit characteristic low-frequency harmonic structures generated by propulsion systems and onboard machinery. These spectral patterns enable CNN architectures to learn discriminative vessel representations directly from acoustic data.

5. LITERATURE REVIEW

Traditional underwater acoustic classification systems primarily relied on handcrafted acoustic descriptors such as Mel-Frequency Cepstral Coefficients (MFCCs), spectral centroid, wavelet features, and zero-crossing rates. These descriptors were commonly combined with classical machine learning algorithms including Logistic Regression, Support Vector Machines (SVMs), and Random Forest classifiers.

Among these approaches, MFCC-based systems became particularly popular because MFCCs effectively capture the spectral envelope characteristics of underwater acoustic signals while maintaining relatively low computational complexity. Classical machine learning models trained on MFCC representations often perform well under limited-data conditions due to the compact and highly informative nature of handcrafted features.

Recent developments in deep learning introduced Convolutional Neural Networks (CNNs) for spectrogram-based acoustic classification. Instead of relying entirely on handcrafted descriptors, CNN architectures automatically learn hierarchical spatial-frequency patterns directly from spectrogram representations. This is particularly useful in underwater acoustics because vessel signatures often contain harmonic structures, cavitation noise, and propeller frequency bands that become visually distinguishable in spectrogram images.

However, deep learning models generally require significantly larger and more balanced datasets to achieve strong generalization performance. Under limited-data conditions, deeper architectures are more susceptible to overfitting, especially in highly imbalanced underwater datasets where dominant vessel categories overshadow minority classes.

6. EXPERIMENTAL SETUP

All experiments were implemented in Python using PyTorch for deep learning pipelines, Scikit-learn for the classical baseline, and Librosa for audio preprocessing and feature extraction. Performance was evaluated primarily using macro-average F1-score, which weights all classes equally regardless of support. Per-class precision, recall, and weighted F1 are additionally reported.

7. IMPLEMENTED MODELS

7.1. Model 1: MFCC + Logistic Regression

The first model served as the classical baseline. MFCC summary statistics were extracted from each segment, standardised using a training-set-fitted scaler, and classified using Logistic Regression with balanced class weighting. This model represents the handcrafted feature engineering approach, where domain knowledge determines the representation prior to classification.

The model used softmax probability outputs for multi-class classification:

$$P(y = i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (2)$$

The model achieved a test accuracy of 70% and a macro F1-score of 0.68. UUV and speed boat classification were strong, achieving F1-scores of 0.96 and 0.88 respectively. Tanker recall was 0.28 — already reflecting the difficulty posed by the under-represented class even with balanced class weighting. Despite its simplicity, the model demonstrated that compact handcrafted features remain competitive for underwater acoustic classification under limited-data conditions.

7.2. Model 2: CNN-Based Spectrogram Classification

The second model replaced handcrafted features with a convolutional neural network trained directly on log-mel spectrograms, enabling automatic spatial-frequency feature extraction. The architecture comprised four convolutional blocks with batch normalisation, max pooling, and dropout, followed by global average pooling and a fully connected classifier. Plain cross-entropy loss was used without explicit minority class handling.

The CNN architecture incorporated convolutional feature extraction, nonlinear activations, validation monitoring, early stopping, and macro-F1 analysis during training. Compared to the baseline model, this approach introduced significantly higher representational capacity and improved spectral pattern learning.

The model achieved a test accuracy of 84% and a macro F1-score of 0.75. Speed boat and UUV classification reached near-perfect scores of 1.00 and 0.99 respectively, and passenger ship recall improved to 0.92. Tanker recall degraded to 0.08 compared to 0.28 in the baseline, highlighting that a CNN trained

on imbalanced data without minority-class handling learns to suppress tanker predictions in favour of majority classes.

The categorical cross-entropy loss function used during CNN training is given by:

$$L = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (3)$$

7.3. Model 3: Sequential Two-Stage Pipeline

The third model introduced a sequential two-stage classification pipeline designed to address the acoustic confusability among large vessel classes and the class imbalance problem. Stage 1 performs coarse five-class classification using an enhanced CNN with squeeze-and-excitation blocks and attention pooling. Samples predicted as large vessels are passed to a dedicated Stage 2 model that resolves the distinction among cargo ships, passenger ships, and tankers. Stage 2 uses a CNN-GRU architecture with three-channel input (log-mel spectrogram, delta, and delta-delta coefficients) and a restricted frequency range of 20–1000 Hz to focus on low-frequency propeller and engine signatures characteristic of large vessels. Both stages employed focal loss with inverse-frequency class weights and a weighted random sampler.

Stage 2 results on large vessels showed cargo ship at F1 0.77 and passenger ship at F1 0.74. Tanker recall improved to 0.19 from 0.08 in Model 2, indicating that isolating large vessel classification into a dedicated focused model provides a measurable benefit. However, the overall Stage 2 macro F1 of 0.55 across the three large vessel classes reflects the continued difficulty of tanker classification.

The t-SNE visualisation of Stage 2 CNN-GRU embeddings (Figure 5) provides further insight. Tanker embeddings form a loose but partially separable cluster, indicating that the model has learned some discriminative structure despite the limited training samples. However, a substantial number of tanker points are scattered across cargo and passenger ship regions, consistent with the low recall observed. Cargo and passenger ship embeddings overlap considerably throughout the space, reflecting their shared low-frequency acoustic characteristics.

The focal loss function is defined as:

$$FL(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

8. RESULTS AND DISCUSSION

8.1. Model 1: MFCC + Logistic Regression

The first model used MFCC feature extraction combined with Logistic Regression classification. The model achieved a test accuracy of 70% and a macro F1-score of 0.68.

TABLE 1: Model 1 Classification Report

Class	Precision	Recall	F1-score
cargo_ship	0.83	0.70	0.76
passenger_ship	0.72	0.69	0.71
speedboat	0.83	0.94	0.88
tanker	0.05	0.28	0.09
unmanned_underwater_vehicle	0.96	0.95	0.96

TABLE 2: Overall Performance of Model 1

Metric	Value
Accuracy	70%
Weighted F1-score	0.73
Macro F1-score	0.68
Support Samples	7121

Strong performance was observed for UUV and speed boat categories, achieving F1-scores of 0.96 and 0.88 respectively. However, tanker classification remained difficult with a recall of 0.28, reflecting the impact of severe class underrepresentation even under the balanced class weighting applied by Logistic Regression.

8.2. Model 2: CNN-Based Spectrogram Classification

The second model explored CNN-based spectrogram classification using log-mel spectrogram representations. The model achieved 84% test accuracy and a macro F1-score of 0.75. Strong performance was observed for speed boats, passenger ships, and UUVs, with near-perfect scores for the latter two.

TABLE 3: Model 2 Classification Report

Class	Precision	Recall	F1-score
cargo_ship	0.93	0.78	0.85
passenger_ship	0.78	0.92	0.84
speed_boat	1.00	1.00	1.00
tanker	0.09	0.08	0.08
uuv	1.00	0.99	0.99

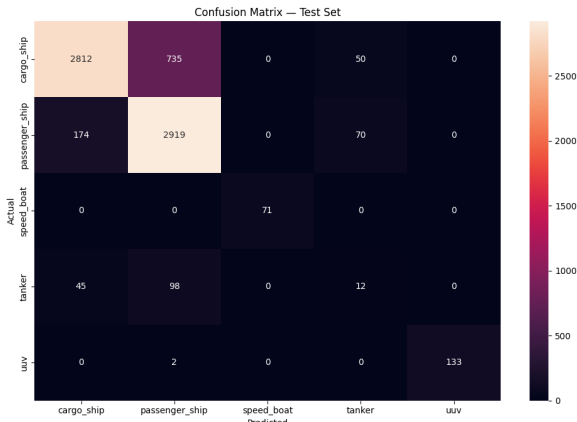


FIGURE 3: Confusion Matrix for Model 2: CNN-Based Spectrogram Classification

TABLE 4: Overall Performance of Model 2

Metric	Value
Accuracy	84%
Weighted F1-score	0.83
Macro F1-score	0.75
Support Samples	7121

However, tanker recall degraded to 0.08 compared to 0.28 in the baseline model, demonstrating that a CNN trained on imbalanced data without explicit minority-class handling learns to suppress predictions for underrepresented classes in favour of dominant ones.

8.3. Model 3: Sequential Two-Stage Pipeline

The final model explored a deeper CNN-based architecture with weighted class handling and staged learning strategies. The model achieved approximately 66% accuracy in stage 1 and 74% accuracy in stage 2.

TABLE 5: Model 3 Classification Report - Stage 2

Class	Precision	Recall	F1-score
cargo_ship	0.81	0.74	0.77
passenger_ship	0.72	0.77	0.74
tanker	0.10	0.19	0.13

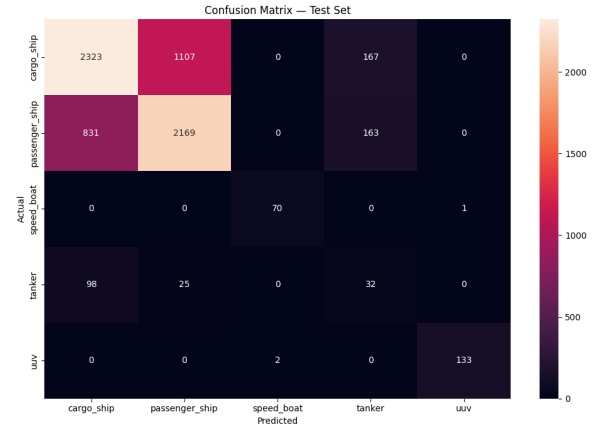


FIGURE 4: Confusion Matrix for Model 3: Sequential Two-Stage Pipeline - Stage 1

TABLE 6: Performance of Model 3 - Stage 2

Metric	Value
Accuracy	74%
Weighted F1-score	0.75
Macro F1-score	0.55
Support Samples	2034

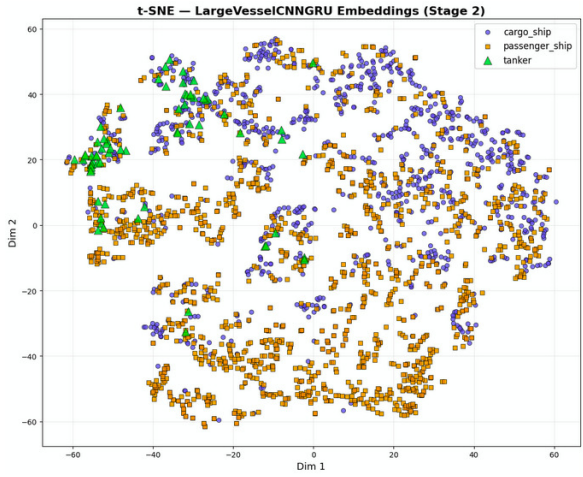


FIGURE 5: t-SNE Plot for Model 3

The deeper architecture achieved relatively stable performance for cargo and passenger vessels. However, tanker classification performance remained poor because of extreme class imbalance and limited training diversity.

8.4. Comparative Analysis

TABLE 7: Comparative Performance of Implemented Models

Model	Feature Type	Accuracy	Macro F1
Model 1	MFCC	70%	0.68
Model 2	Log Mel Spectrogram	84%	0.75
Model 3	Spectrogram - Stage 2	74%	0.55

The baseline MFCC + Logistic Regression model achieved competitive performance despite its simplicity, demonstrating the effectiveness of handcrafted spectral descriptors under limited-data conditions. Notably, it achieved the highest tanker recall of 0.28 across all three models, suggesting that the balanced class weighting applied in Logistic Regression provided more consistent minority-class coverage than the deeper models trained on imbalanced data.

The CNN-based Model 2 achieved the strongest overall performance with 84% test accuracy and a macro F1-score of 0.75, with near-perfect classification for speed boats and UUVs. However, tanker recall degraded to 0.08, illustrating that increased model capacity without explicit minority-class handling does not translate to improved minority-class performance.

The sequential two-stage pipeline partially recovered tanker recall to 0.19 through the combination of focal loss, weighted random sampling, and a dedicated large-vessel model. Despite this improvement, tanker performance remained limited across all models. The macro F1 scores of Model 2 and Model 3 being comparable indicates that the additional architectural complexity of the sequential pipeline did not yield a substantial overall gain, further supporting the conclusion that data scarcity rather than model design is the primary limiting factor.

9. DISCUSSION

The experiments demonstrate that underwater acoustic signals contain sufficient spectral information for machine learning-based vessel classification across both classical and deep learning paradigms. The baseline Logistic Regression model proved competitive under constrained data conditions, while CNN-based architectures successfully captured spatial-frequency structures difficult to represent using fixed descriptors.

The central finding is that class imbalance — particularly the severe underrepresentation of the tanker class — is the primary bottleneck, not model capacity or architectural choice. The similarity in macro F1 between Model 2 and Model 3 despite the additional complexity of the sequential pipeline reinforces this conclusion. The performance gap between the primary dataset results and DS3500 provides direct empirical support.

10. FUTURE WORK

The most direct avenue for improvement is the acquisition of additional tanker recordings. However, the embedding analysis suggests that increasing tanker sample count alone may not fully resolve the classification difficulty, as substantial overlap exists between tanker, cargo ship, and passenger ship acoustic representations. Future work should therefore investigate methods capable of learning more discriminative temporal and spectral representations for acoustically similar vessel classes.

Improved class balancing strategies, transfer learning using pretrained audio models, and hybrid CNN-RNN architectures may also improve minority-class discrimination and overall generalization performance. Real-time underwater deployment and streaming acoustic analysis remain promising long-term extensions of this work.

11. CONCLUSION

This project investigated underwater vessel classification using passive acoustic sensing across three machine learning approaches: an MFCC-based Logistic Regression baseline, a CNN-based spectrogram classifier, and a sequential two-stage pipeline with dedicated large-vessel discrimination.

The MFCC + Logistic Regression baseline achieved a test accuracy of 70% and a macro F1-score of 0.68, demonstrating the competitiveness of handcrafted spectral features under limited-data conditions. The CNN-based spectrogram models explored automatic feature extraction and nonlinear representation learning, with Model 2 achieving the strongest overall performance of 84% test accuracy and a macro F1-score of 0.75. Despite additional mitigation strategies, the sequential two-stage pipeline achieved a comparable macro F1-score, indicating that data scarcity rather than model architecture is the primary limiting factor.

The experiments highlight an important practical observation: class imbalance, particularly the severe underrepresentation of the tanker class, is the primary bottleneck to minority-class recall. This was directly confirmed by the 0.86 macro F1-score achieved on the balanced DS3500 dataset, demonstrating that the proposed architectures are capable of strong performance when training data is sufficient and balanced.

Overall, the project demonstrates the feasibility of passive acoustic vessel classification for maritime surveillance, underwater monitoring, and naval security applications.

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